

In[363]:=

```
ClearAll["Global`*"];
清除全部

SeedRandom[];
随机种子

(* ===== *)
(*1. 并发暴胀引擎 (完全原生 Union 去重, 无人工排序干预)*)
并集

(* ===== *)
StepEvolution[g_Graph, window_] := Module[
图 模块
{allEdges, activeEdges, nActive, usedEdges, updates, tempE1, tempE2, x, y, z, wCounter,
newVertices, newActive, newFrozen, remainingEdges}, allEdges = EdgeList[g];
边的列表

activeEdges = RandomSample[Cases[allEdges, _UndirectedEdge]];
伪随机采样 模式匹配 无向边

nActive = Length[activeEdges];
长度

If[nActive < 2, Return[{g, 0}]];
如果 返回

usedEdges = {}; updates = {};
Do[tempE1 = activeEdges[[i]];
Do循环

If[! MemberQ[usedEdges, tempE1], Do[tempE2 = activeEdges[[j]];
如果 成员判定 Do循环

If[! MemberQ[usedEdges, tempE2] &&
如果 成员判定

Length[Intersection[List @@ tempE1, List @@ tempE2]] == 1,
长度 交集 列表 列表

AppendTo[updates, {tempE1, tempE2}];
附加

AppendTo[usedEdges, tempE1]; AppendTo[usedEdges, tempE2];
附加 附加

Break[]];, {j, i + 1, Min[i + window, nActive] }], {i, 1, nActive}];
跳出循环 最小值

If[Length[updates] == 0, Return[{g, 0}]];
返回 长度 返回

wCounter = Max[VertexList[g]];
顶点列表

newVertices = {}; newActive = {}; newFrozen = {};
Do[y = Intersection[List @@ pair[[1]], List @@ pair[[2]] [[1]];
Do循环 交集 列表 列表

x = Complement[List @@ pair[[1]], {y}] [[1]];
补集 列表

z = Complement[List @@ pair[[2]], {y}] [[1]];
补集 列表

wCounter++;
AppendTo[newVertices, wCounter];
附加

newActive = Join[newActive, {UndirectedEdge[x, z],
连接 无向边

UndirectedEdge[x, wCounter], UndirectedEdge[wCounter, z]}];
无向边 无向边
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newFrozen = Join[newFrozen, {DirectedEdge[x, y], DirectedEdge[y, z]}];
      连接      有向边      有向边
{pair, updates}];
remainingEdges = Complement[allEdges, Flatten[updates]];
      补集      压平
{Graph[Union[Join[VertexList[g], newVertices]],
  图      并集      连接      顶点列表
    Union[Join[remainingEdges, newActive, newFrozen]]], Length[updates]]];
      并集      连接      长度

(* =====*)
(*2. 并发宇宙演化采集 (40 步)*)
(* =====*)
Print[Style["[1/2] 正在点燃并发暴胀宇宙... (目标: 40步)", Blue, Bold]];
打印      样式      蓝色      粗体
universe = CompleteGraph[6];
      完全图
steps = 100;
widthData = {};

Monitor[Do[{universe, concurrentCount} = StepEvolution[universe, 50];
  监控      Do循环
    If[concurrentCount > 0,
      如果
        activeNodes = Union[Flatten[List@@@ Cases[EdgeList[universe], _UndirectedEdge]]];
          并集      压平      列表      模...      边的列表      无向边
        If[Length[activeNodes] > 1, dists = (GraphDistance[universe, 1, #] & /@ activeNodes);
          ...      长度      图中顶点的距离
          AppendTo[widthData, {i, StandardDeviation[Select[dists, NumericQ]]}];];];
          附加      标准偏差      选择      数值量判定
        {i, 1, steps}], ProgressIndicator[i, {0, steps}]];
          进度指示器

(* =====*)
(*3. TCCT 全息肖尔分析 (Topological Sholl Analysis)*)
(* =====*)
Print[Style["\n[2/2] 演化停止, 启动全息肖尔形态学分析...", Purple, Bold, 14]];
打印      样式      紫色      粗体

(*强制将当前宇宙退化为 4D 纯因果连通图*)
      偏导
undirectedUniverse =
  Graph[VertexList[universe], UndirectedEdge@@@ EdgeList[universe]];
      图      顶点列表      无向边      边的列表
seedNode = 1; (*1 号节点作为大爆炸原初奇点*)

(*发射全局测地线*)
distances = GraphDistance[undirectedUniverse, seedNode];
      图中顶点的距离
shellCounts = Counts[distances];
      关联计数
shollData = Sort@KeyValueMap[List, shellCounts];
      排序      键值映射      列表

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radii = shollData[[All, 1]];
volumes = shollData[[All, 2]];

(*提取核心拓扑参量*)
shollCriticalValue = Max[volumes];
shollCriticalRadius = radii[[First[Position[volumes, shollCriticalValue]]][1]];
primaryBranches = VertexDegree[undirectedUniverse, seedNode];
ramificationIndex = N[shollCriticalValue / primaryBranches];

decayData = Select[shollData, #[[1]] > shollCriticalRadius &];
If[Length[decayData] > 2, logDecayData = {#[[1]], Log[#[[2]]]} & /@ decayData;
fitDecay = LinearModelFit[logDecayData, x, x];
shollDecayK = fitDecay["BestFitParameters"][[2]];
r2Decay = fitDecay["RSquared"]; shollDecayK = 0; r2Decay = 0;];

(*绘制:肖尔宇宙剖面图*)
shollPlot = Show[ListLinePlot[shollData, PlotStyle → Directive[Thick, Darker[Cyan]],
Filling → Axis, FillingStyle → Opacity[0.2, Cyan], PlotTheme → "Detailed",
ImageSize → 600, FrameLabel → {Style["Topological Radius (r)", 12, Bold],
Style["Spatial Volume V(r)", 12, Bold]},
PlotLabel → Style["Cosmological Sholl Profile", 14, Bold]], Graphics[
{{Red, PointSize[0.03], Point[{0, shollData[[1, 2]]}], {Purple, Dashed, Thick, Line[
{{shollCriticalRadius, 0}, {shollCriticalRadius, shollCriticalValue}}}}}}];

Print[shollPlot];

(*输出物理张量表*)
shollTable = {{"Sholl Metric (Neuro-Morphology)",
"Cosmological Physical Mapping", "Extracted Value"},
{"Primary Branches", "Big Bang Initial Degrees of Freedom", primaryBranches},
{"Sholl Critical Radius (r_c)", "Turnaround Time (Expansion to Collapse)",
shollCriticalRadius}, {"Sholl Critical Value (N_max)",
"Maximum Equator Spatial Volume", shollCriticalValue}, {"Ramification Index",
"Inflation Amplification Factor", NumberForm[ramificationIndex, {Infinity, 2}]},
{"Sholl Regression Coeff (k)", "Big Crunch Topological Deceleration",

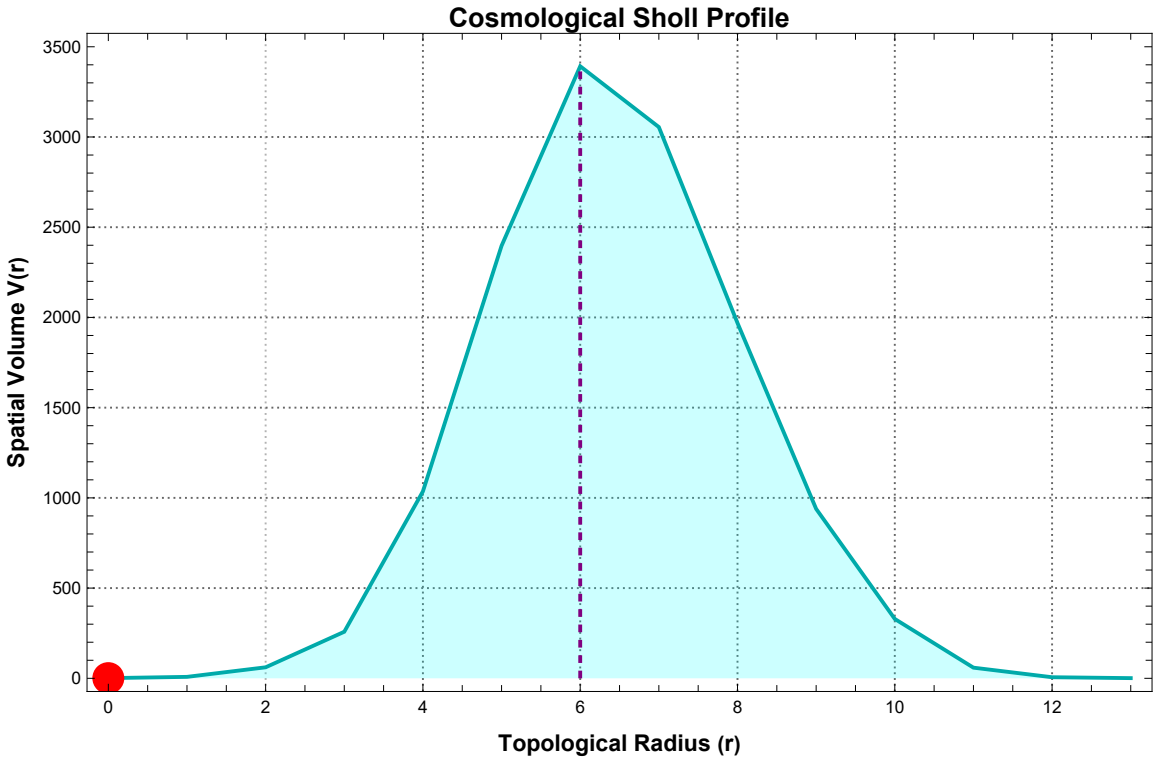
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If[shollDecayK ≠ 0, ToString[NumberForm[shollDecayK, {Infinity, 4}]] <>
  " (R^2=" <> ToString[Round[r2Decay, 0.01]] <> ") ", "N/A"]];

Print[Grid[shollTable, Frame → All, Background → {None, {LightGray, None}},
  Alignment → Left, ItemSize → {{22, 28, 15}, Automatic}]];

[1/2] 正在点燃并发暴胀宇宙... (目标: 40步)
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[2/2] 演化停止, 启动全息肖尔形态学分析...



Sholl Metric (Neuro-Morphology)	Cosmological Physical Mapping	Ext
Primary Branches	Big Bang Initial Degrees of Freedom	10
Sholl Critical Radius (r_c)	Turnaround Time (Expansion to Collapse)	6
Sholl Critical Value (N_{max})	Maximum Equator Spatial Volume	339
Ramification Index	Inflation Amplification Factor	339
Sholl Regression Coeff (k)	Big Crunch Topological Deceleration	-1